Advanced Machine Learning

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April 9, 2026

Contents

1 Monday April 6th

1.1 Introduction

An introduction was provided for statistical learning, however I neglected to take notes.

1.2 Data Types

There are different paradigms for types of data.
For qualitative of categorical data, we have nominal and ordinal data. Nominal describes data that can be lumped into categories with no numerical relation to each other. For example, the category of country has Albania, Greece, Russia, etc. Nominal implies there is no ordering between these categories, whereas ordinal data describes data that has an inherent size or ordering to it. For example, a "size" category where values can include small, medium, or large. These are categorical data entries describing qualitative features of the data, however there are direct implications to the size and relation of one category to the other, therefore it is ordinal.
For quantative data, we have discrete and continuous data types.

1.3 Learning Types

Unsupervised, supervised, and semi-supervised.

2 Tuesday April 7th

2.1 Gaussian Distributions

An $\mathbb{R}^d$-valued random vector $\mathbf{X} = [X_1, \ldots, X_d]^\top$ is said to be Gaussian, if for any fixed vector $\mathbf{a} = [a_1, \ldots, a_d]^\top \in \mathbb{R}^d$, the real r.v. $a_1X_1 + \cdots + a_dX_d$ (the dot product) is Gaussian.
In the Gaussian case, a zero covariance between parts of a random variable means that the values are also independent. This is not the case in general for all distributions.

2.2 Monte Carlo Method

Considering a continuous function h mapping inputs between 0 and 1 to any real number

$$I = \int_0^1 h(x)dx = \int_{\mathbb{R}} h(x)\mathbf{1}_{[0,1]}(x)dx = E[h(X)] \tag{1}$$

where X is a random variable following a [0,1] uniform distribution. Here, we transform a deterministic problem into a stochastic one.
We can then calculate the expectation by collecting samples X_i and using the mean as an approximation for the true expectation.

$$\frac{h(X_1) + h(X_2) + \cdots + h(X_n)}{n} \approx E(h(X)) \tag{2}$$

Monte Carlo relies on a random generation of a uniform distribution from [0,1].
Weibull example:

$$y = F(x) = 1 - e^{(-x\alpha)^\beta}, x \geq 0 \tag{3}$$
$$y = F^{-1}(y) = \alpha[-ln(1-y)]^{1/\beta} \tag{4}$$
$$\tilde{X} = \alpha[-ln(1-U)]^{1/\beta} \tag{5}$$
$$\tilde{X} = \alpha[-ln(U)]^{1/\beta} \tag{6}$$

Performing algorithm:

$$\textbf{Step 1:} \quad u\ U(0,1) \tag{7}$$
$$\textbf{Step 2:} \quad X = [-ln(u)]^{2/3} \tag{8}$$

2.2.1 Precision

The error of the estimate is

$$R_n = \frac{h(X_1) + \cdots + h(X_n)}{n} - E[h(X)] \tag{9}$$

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We define the precision by parameters ε, θ, which values are small, as follows

$$P(|R_n| \leq \varepsilon) \geq 1 - \theta \tag{10}$$

or

$$P(|R_n| \geq \varepsilon) \leq \theta \tag{11}$$

where in order to maintain a set level of precision, we have certain constraints on the variance bound by our number of samples and the set values of ε, θ.

$$Var(\hat{\gamma}_n) = \frac{1}{n}\{E[h(Y_1)(\frac{f(Y_1)}{g(Y_1)})^2] - \gamma^2\} \tag{12}$$

2.3 Bootstrapping

$$\theta = \mathbb{P}(X \leq 1) = e^{-\lambda}(1 + \lambda) \tag{13}$$

3 Wednesday April 8th

3.1 Simple Linear Regression

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})y_i}{\sum_{i=1}^n (x_1 - \bar{x})^2} \tag{14}$$

$$\hat{\beta}_1 \backsim N(\beta_1, \frac{\sigma^2}{S_{xx}}) \tag{15}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \tag{16}$$

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \tag{17}$$

Michaelis-Menten model:

$$y_i = \mathbf{E}(Y_i) = \frac{\theta_1 x_i}{x_i + \theta_2} \tag{18}$$

proving $cov(\hat{b}_n, \bar{y}) = 0$

$$Cov(\hat{b}_n, \bar{y}) = \frac{1}{n}\sum_{i=1}^{n}(\frac{S_{xy}}{S_{xx}} - \bar{b}_n)(\bar{y} - \bar{y}) = 0 \tag{19}$$

3.2 Multiple Linear Regression

4 Thursday April 9th

4.1 Principal Component Analysis